Aaryan Patel

Menlo Park, CA | (437) 443-2790 | aaryanp35@gmail.com | Portfolio

PROFILE

Second-year University of Toronto student-athlete majoring in Economics with Data Analytics (minors: Computer Science, Statistics). Combines rigorous quantitative training with hands-on experience building data pipelines and predictive models. Proven ability to perform under pressure and manage competing demands, balancing varsity athletics with a demanding academic program. Eager to apply analytical and technical skills in a fast-paced professional environment.

EDUCATION

University of Toronto | Economics with Data Analytics | Minors: CS & Statistics *2024 – Present*

- Relevant courses: Advanced Microeconomic Theory, Advanced Macroeconomic Theory, Probability & Statistics I, Linear Algebra, Big-Data Tools for Economists, Foundations of CS I & II, Software Design.
- Varsity Student-Athlete (Rowing)

Menlo Atherton High School | High School Diploma *2020 – 2023*

- Founded MA Hackers Club (2023); completed AP CS A & Principles and college-level Economics.

RESEARCH EXPERIENCE & PROJECTS

Independent Project | Proxii — Spatial Economics & Gentrification Risk Platform *Spring 2026*

- Designed a spatial economics research tool that ingests real-time business data via the Google Maps Places API to model neighborhood economic transitions
- Constructed a "Proxii Score" index to quantify gentrification risk from point-of-interest density and sentiment signals (e.g., specialty coffee shops, fitness studios), applying principles of hedonic pricing and spatial econometrics.
- Built end-to-end data pipelines in Python and SQL to clean, merge, and model large-scale location datasets;

Independent Project | Music Mood Classifier — ML Feature Engineering *Spring 2025*

- Trained a decision-tree classifier on six Spotify audio features (danceability, valence, energy, etc.) to predict a song for the listener based on user mood, demonstrating proficiency in supervised learning and feature engineering.
- Extracted and processed audio attribute data programmatically via the Spotify API; designed a replicable pipeline applicable to broader consumer-behavior research.

University of Toronto | In Progress | Neighborhood Home Values & Police Search Outcomes — ECO225 Research Paper *Spring 2026*

- Investigating whether neighborhood home values predict the likelihood a police stop results in a search, using low-level/misdemeanor arrest rates as a mediating mechanism (X→Z→Y causal framework); merged 250,093 SFPD stops with Zillow Home Value Index data at the neighborhood level.
- Built eight OLS regression specifications with HC1 robust standard errors (linear, log-linear, interaction, and subgroup models); produced choropleth maps and scatter plots visualizing spatial enforcement patterns across San Francisco neighborhoods.
- Extended analysis with HTML-based web scraping to supplement neighborhood-level data, and implemented regression tree and random forest models to capture non-linear enforcement patterns not detectable via OLS.
- Found that home values predict search likelihood, but the effect is largely mediated by low-level stop concentration, consistent with order-maintenance policing theory (Harcourt 2001) and Beck's (2020) development-directed enforcement framework.

PROFESSIONAL EXPERIENCE

American Precision Gear | Engineering Intern *Summer 2024*

- Maintained manufacturing standards by executing quality control and inspection processes for critical aircraft and space vehicle parts.
- Gained hands-on technical experience in the machine shop, working directly with equipment such as mills, routers, and cutters.
- Managed logistics and supply chain functions, overseeing secure shipping and packing procedures to ensure safe delivery of finished components to clients.

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, Scikit-learn), SQL, MATLAB, Java, C++

Econometrics & ML: OLS/IV Regression, Logistic Regression, Classification & Regression Trees, Random Forest, Clustering, Time Series Analysis

Data Tools: Matplotlib, Plotly, Jupyter Notebooks, Git, Google Cloud Platform (GCP), AWS

Research Methods: Data pipeline construction, API data collection, feature engineering, predictive modeling, data visualization

LEADERSHIP & OUTREACH

University of Toronto / Norcal Rowing Crew | Varsity Rower & Team Captain *2020 – Present*

- Competing as a Division I-level student-athlete at UofT while maintaining a full academic course load; demonstrating sustained discipline, time management, and commitment to excellence in two demanding areas simultaneously.
- Captained Norcal Crew (250+ athletes) to multiple national championships; selected for U.S. Rowing Olympic Development Program, one of the most competitive selection processes in youth rowing.

Community STEM Outreach | Lead Coordinator & Guest Speaker *2023 – 2024*

- Designed and delivered computer literacy and coding curricula for underserved communities and seniors; presented to audiences of 50+, bridging the digital divide through mentorship.

AWARDS & RECOGNITION

College Board National African American & Hispanic Recognition Award (2024) • Norcal Rowing: Leadership Award (2024), Best Teammate (2023), Hardest Worker (2021)